Hellenic Air Force Academy

Department of Aeronautical Sciences

Division: Electronics, Electrical Power, Telecommunications

Diploma Thesis

"Recognition and automatic calculation of the number of people entering a building"

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Supervisor: Assistant Professor Pavlatos Christos

What is the issue that concerns the παγκόσμια community in recent 2 years"

- Detection on February 26, 2020 in Greece
- More than 3,450,000 cases
- More than 29,500 human lives lost

- Measures of protection
- In our daily life
-

Impacts on the Armed Forces

Protection measures

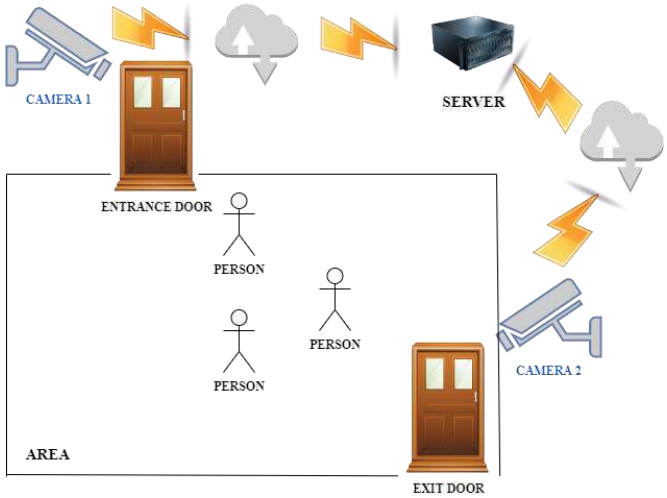
- in the work

Isolation of contacts

- (lectures, briefings, meetings).

ANSC: To contribute to the fight against the coronavirus, by avoiding crowding of staff, by controlling it in enclosed spaces, using simple technological means.

Achieving our goal will provide facilitation, accuracy, and release of involved staff.



DESCRIPTION OF THE PRESENTATION

1. Understanding Program Architecture

2. Theoretical Background

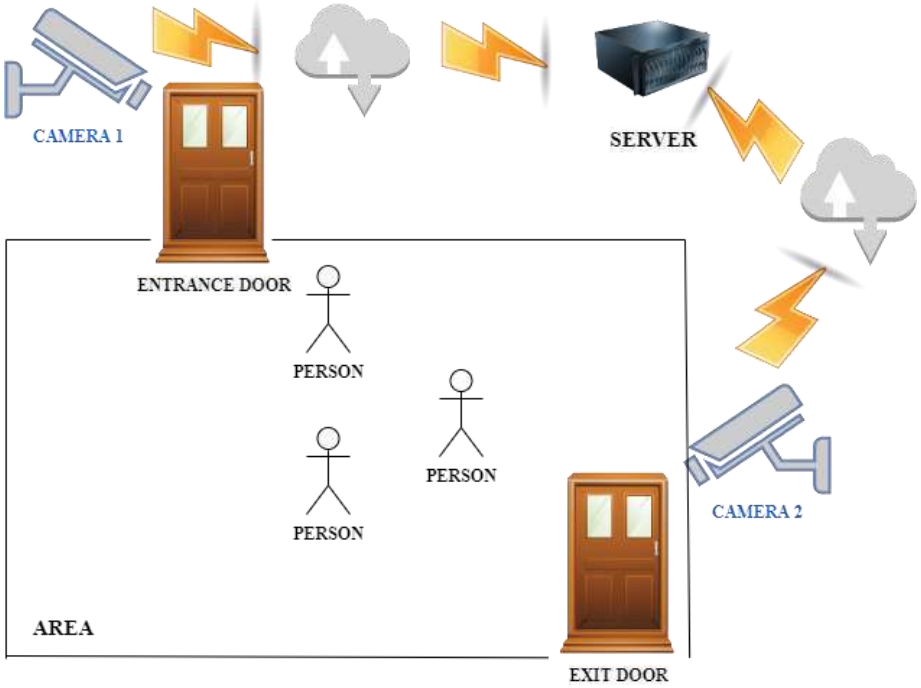
3. Materials and Media

4. Software

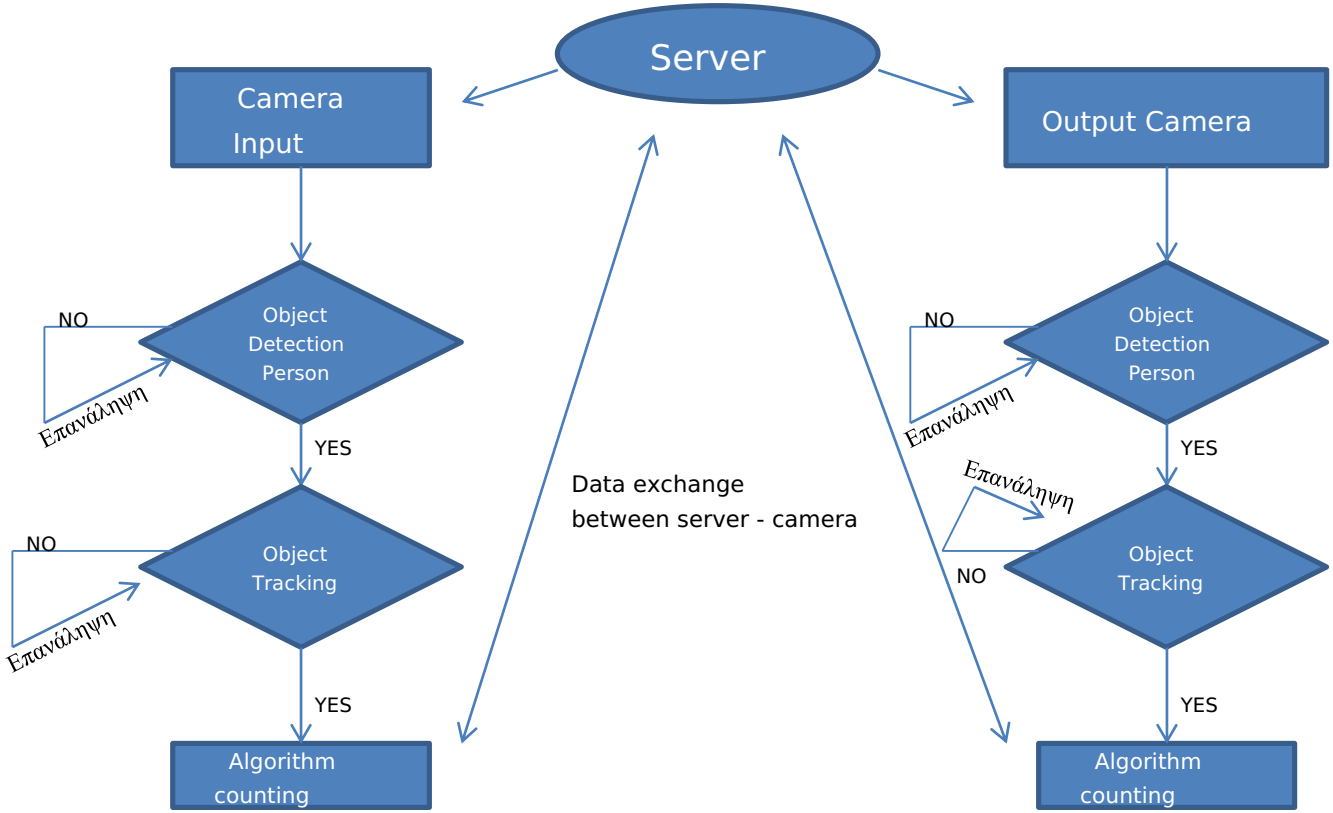
5. Final Outcome

The presentation will last 20 minutes.

1. Understanding Program Architecture



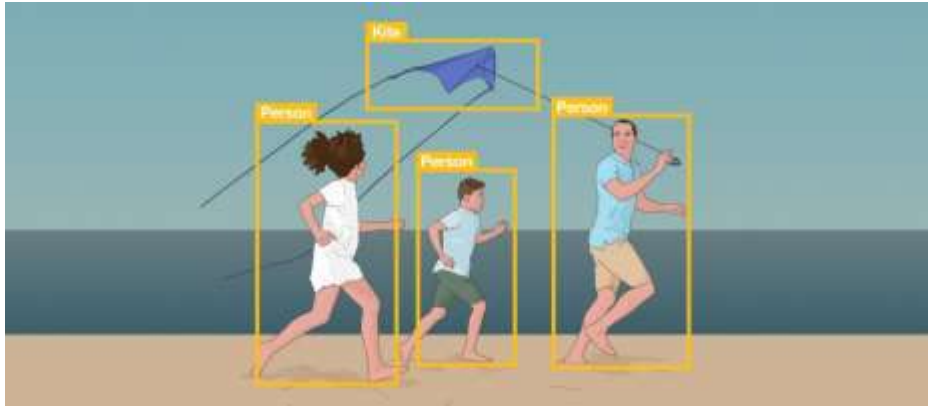
Flow Diagram

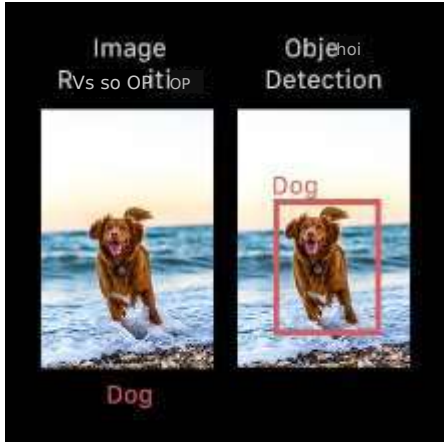


2. Theoretical Background

- Object Detection

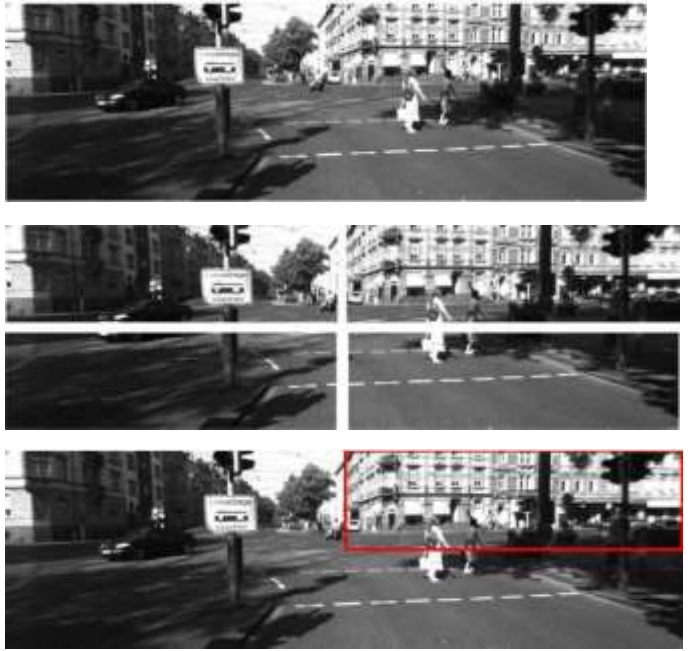
Object detection (Object Detection) is a computer vision technique that works to locate objects within an image/video.



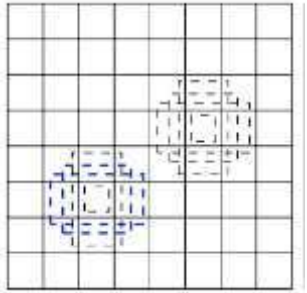


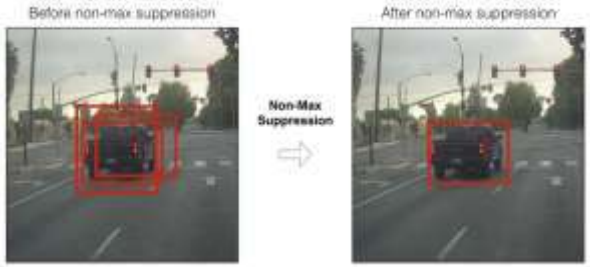
• Object Detection Algorithms

Today there are various models for achieving Object Detection, with the most popular being Haar Cascades and Deep Learning based algorithms such as the Single Shot Detector and the family of R-CNN (R-CNN, Fast R-CNN, Faster R-CNN).



Simple way to perform Object Detection using Deep Learning





Single Shot Detector - Single Shot Detector

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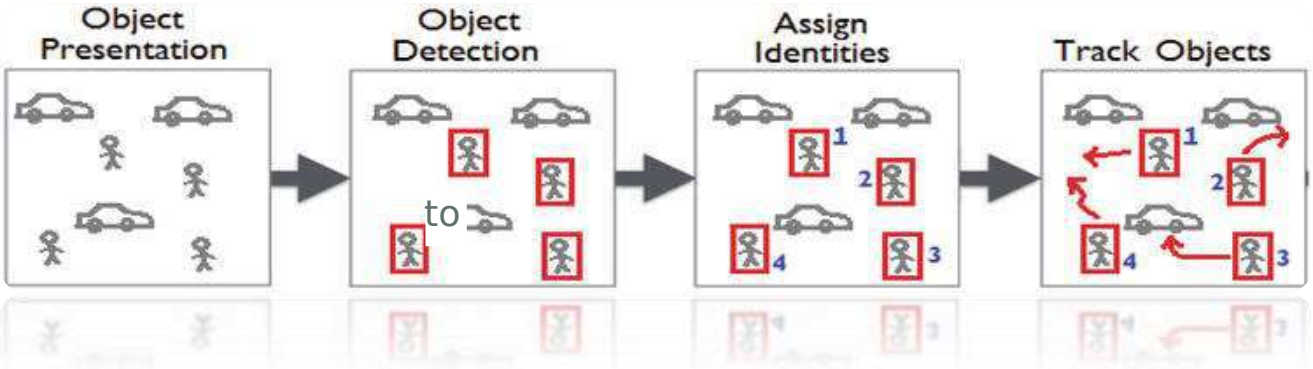
- Object Tracking

Object tracking (Object Tracking) is a process of monitoring the movement of one or more objects, through estimation or prediction of the positions of moving objects in a video.



- Object Tracking Algorithms

There are two types of Object Tracking :

1) Tracking a single object (SOT)

2) Tracking multiple objects (MOT).

- OpenCV Object Tracking

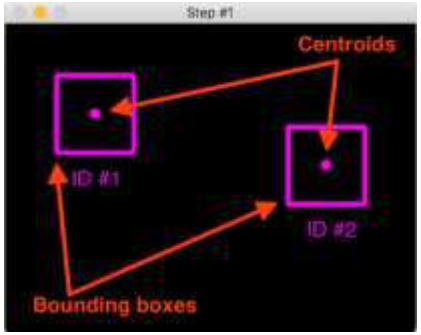
It is a popular and reliable method, with many built-in algorithms, specifically optimized for the needs and goals of object tracking or motion tracking.

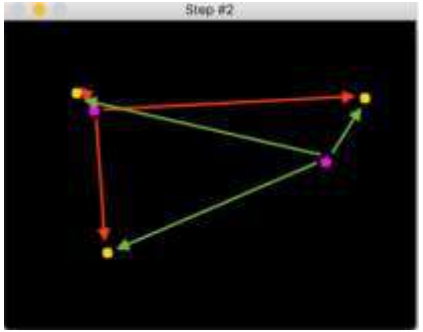
- DeepSORT

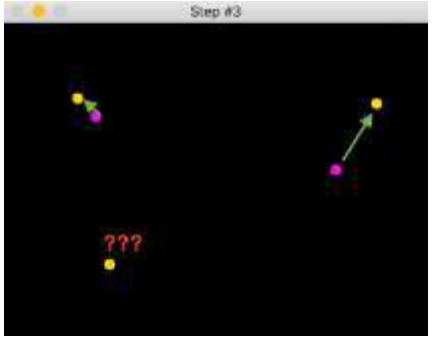
- MDNet

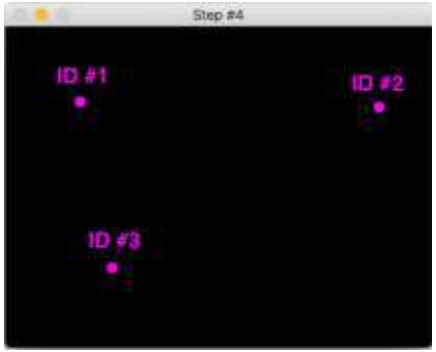
- Object Tracking MATLAB

- Centroid Tracker





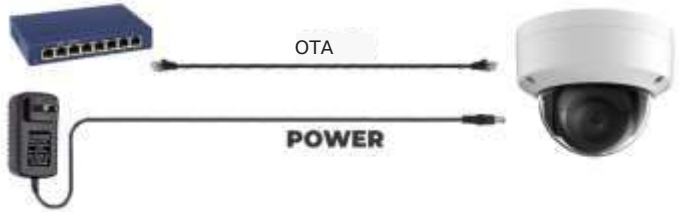




3. Materials and Media

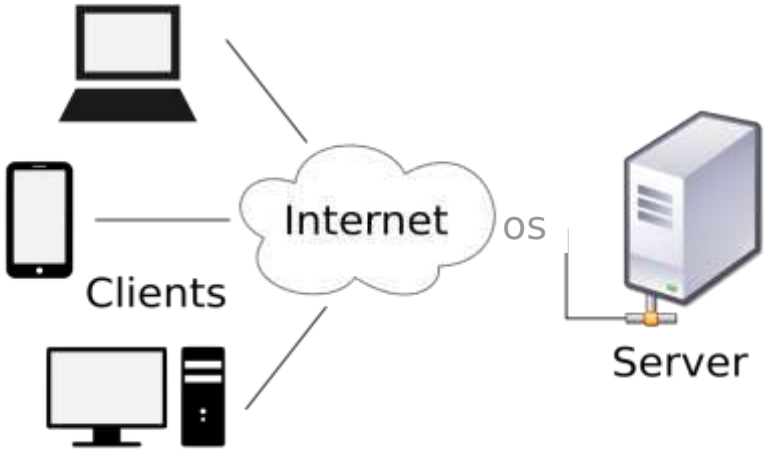
- IP Camera

H IP Camera, is a type of digital video camera that sends image data over an IP network (Internet Protocol).



• Server

It is a piece of computer hardware or software that provides functionality for other programs or devices, which are called customers (customers).



- Raspberry Pi

Initially the program was developed on the Raspberry Pi, which is a fully equipped and functional computer that occupies only the size of a credit card.

Unfortunately it proved to be an inappropriate medium due to low accuracy and performance.

Raspberry Pi

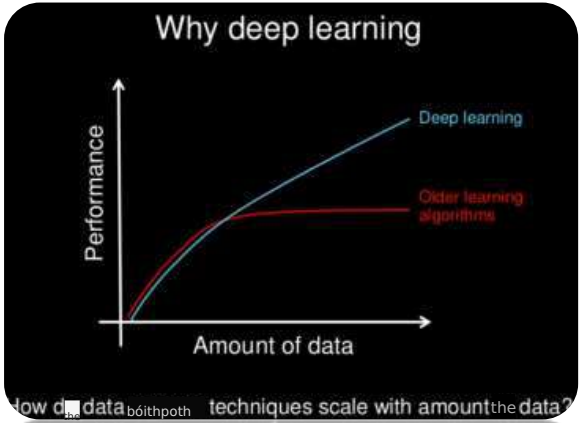
4. Software

The program uses Deep Learning as a base, with main tools:

1) Object Detection

2) Object Tracking

Deep Learning is a set of algorithms that mimics the way humans acquire certain types of knowledge.



Mode of operation

It is equipped with training data and uses the information it receives to create a feature set and build a predictive model.

The training process of a model is time-consuming and complex => a pre-trained caffemodel that recognizes the human form and movement.

Operating system: Linux: (Lightweight, secure, efficient)

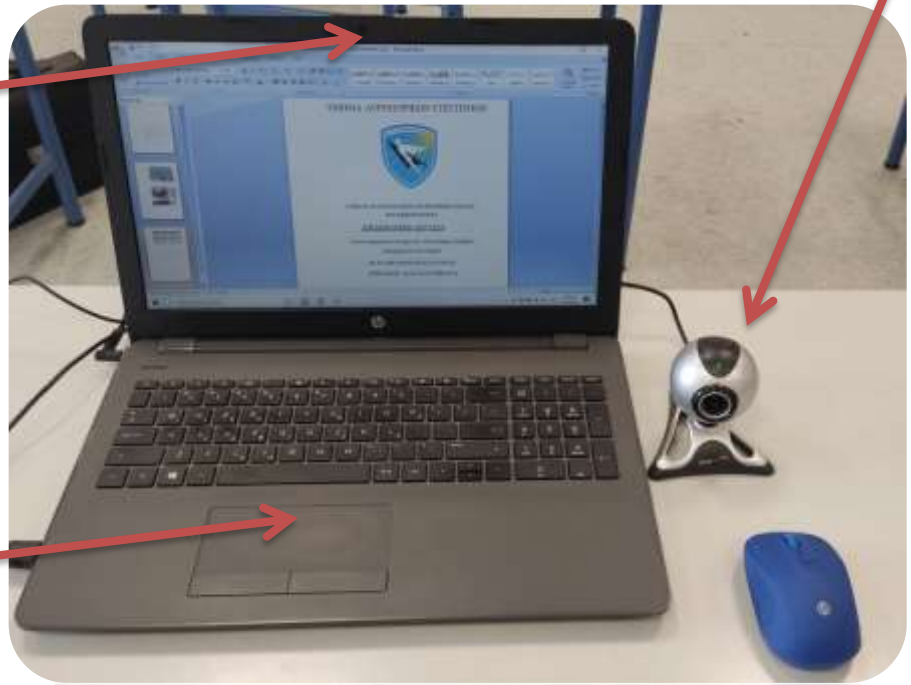
Programming language: Python (Easy, simple structure – syntax, speed)

Program development: Thonny (integrated development environment for Python designed for beginners)

5. Final Result

Camera 2

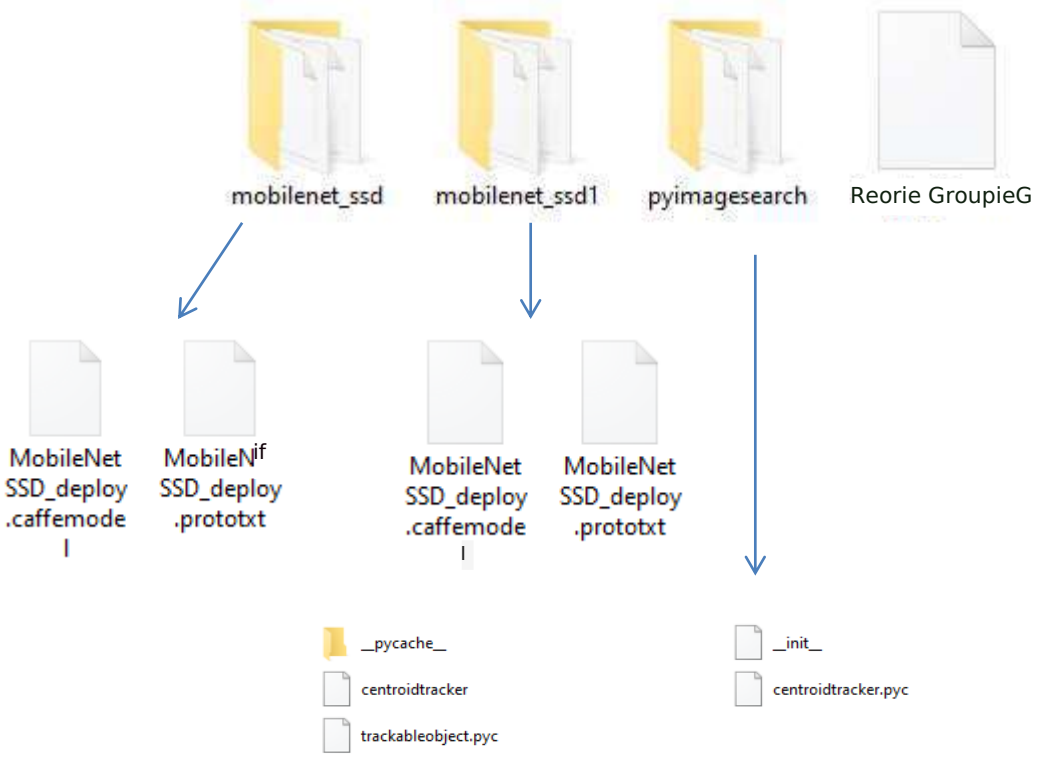
Camera 1

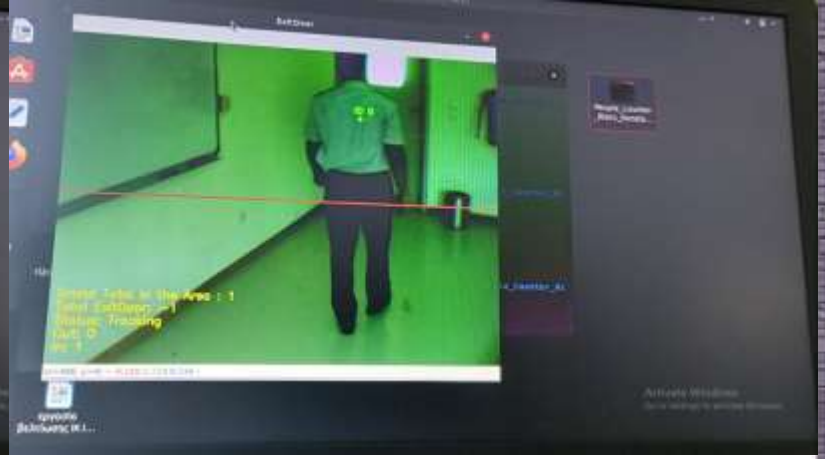


Server

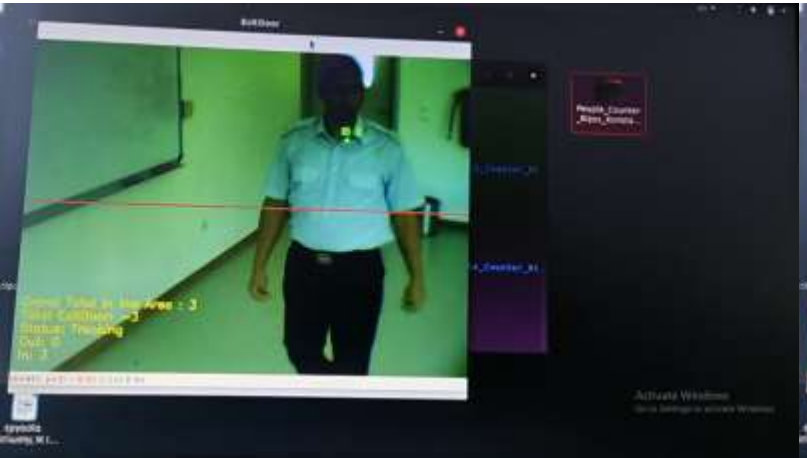
Program Files Analysis

People_Counter_Rizos_Konstantinos





Example of EXIT CAMERA



Technology makes possibilities.
Design makes solutions. Avì makes
questions. Leadership makes actions.

John Maeda

quotefancy

Thank you very much for your attention

A demonstration of the program will follow

I am at your disposal for any questions